

Research Article

Navigating Immigration as an Alien: A Critical Interface Analysis of the US Citizenship and Immigration Services Website

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Abstract—Introduction: This article provides a critical interface analysis of the US Citizenship and Immigration Services (USCIS) website to reveal how systemic oppressions embedded in governmental websites create injustice among minoritized communities. **Literature review:** We situate this research within the existing scholarship about the multilingual user interface, usability studies, and the issues of linguistic social justice as it intersects with technical and professional communication. **Research questions:** 1. How does the USCIS website's content cater to diverse immigrant populations in terms of usability, specifically considering Web Content Accessibility Guidelines (WCAG) for accessible websites? 2. From a user-experience perspective, are the USCIS website navigation tools obtrusive in presenting information? Are there issues of power and privilege through the inclusion/exclusion of certain voices? 3. What ideological and cultural assumptions does its interface design impart to diverse website users through its tools, content organization logic, and visual style? **Research method:** Working under the critical interface analysis framework and adopting a walkthrough approach, we analyze the official website of the USCIS. **Results/discussion:** The USCIS website prioritizes English or Spanish proficiency, potentially excluding users with other diverse language backgrounds. First-time users lack immediate access to essential features, and the site overlooks the needs of its diverse immigrant population, with limited language options, multimedia resources, and occasional discrepancies in content. Using terms like “alien” contradicts the inclusive image the US aims for. The Multilingual Resource Center faces document translation shortages, contributing to a potential digital divide. Inclusive design choices are crucial for creating a welcoming environment and addressing these concerns. **Conclusion:** These findings have implications for understanding the rhetorics of immigration policy, power, identity, and government perceptions.

Index Terms—Critical interface analysis, immigrant users, usability studies.

Technological interfaces function rhetorically to “create various degrees of interaction and engagement” and work as a “central place of interaction for the technological, human, social, and cultural aspects” [1, p. 164]. However, when interfaces are designed without the user's background, culture, and literacy in mind, it creates an issue with the usability of any interface. When user inputs are not considered, such interface design reflects the locus of power, the social and political contexts, and the empowerment and disempowerment of certain voices.

For example, the interface at the end of your journey in the well-known Uber ride-sharing application encourages you to add a tip and cannot be closed until you do so. This hinders people who want to give cash tips. It also frustrates people who have valid reasons for not tipping. Acharya defines usability as

a quality or characteristic of a product to determine whether or not the product is efficient, effective, and satisfying for those who use it. [2, p.7]

Based on this definition, the Uber app is neither efficient nor effective. In addition, its technological affordances may favor one mode of interactivity (tipping by using the payment preferred by Uber while constraining another, such as cash tipping or no tipping) [3].

Sano-Franchini [4] suggested that user-experience (UX) designers must be attentive to how technology designs create new relational circuits and how those relationships work to mediate the dissemination of politicized content. For example, on X (formerly Twitter), such relational circuits can

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be seen in the dissemination of misinformation regarding COVID-19 vaccines [5]. Similarly, the US Citizenship and Immigration Services (USCIS) website [6], the focus of this study, often lacks sufficient consideration for its primary users, who are predominantly refugees and immigrants.

The USCIS website is the first stop for a refugee or immigrant to understand what is next in their immigration journey. That journey can range from getting a work permit to information about legal status to becoming a US citizen. When the pragmatic and functional goals-related aspects of usability, such as learnability, memorability, error recovery, utility, and time as discussed by Nielsen [7] and Quesenberry [8], are not taken into consideration, it hinders access to information. Despite the USCIS website offering numerous content and various navigation tools to help refugees and immigrants resettle in the US, its interface design, organization of tools and content, and the rhetorical use of language reflect systemic inequities [9].

At first, the website is intimidating with hundreds of clickable headings and subheadings, a virtual agent, a variety of lists, and so on. When a person lacking specific legal knowledge, information, literacy, and context of the US attempts to navigate the website, it becomes challenging. They may encounter usability issues and face potential failure in accessing essential information. As such, the affective and intentional interface design affects how minority populations understand their rights

to live and work in the US [10], [11]. Responding to the call for usability that focuses on social justice and digital interface analysis, we present a critical interface analysis [2], [3], [12], [13], [14], [15], [16] of the USCIS website. This analysis reveals how systemic oppressions embedded in governmental websites create injustice among minority communities. We argue that websites like USCIS should be more inclusive, addressing the needs of multilingual, multicultural users who may have lower literacy levels.

Studying the interface design of a website like USCIS is important for two main reasons. First, from a usability perspective, it helps us assess whether various users (particularly refugees and immigrants) can access the necessary information. Second, from a social justice perspective, it is crucial to uncover the ideological biases in the US government's approach toward the minority population. Analyzing the interface design can make a case for why there is a need for a more equitable design that serves the diverse immigrant population fairly. The US has a substantial refugee and immigrant population. According to the Congressional Budget Office, "about 45 million people living in the United States in 2021 were born in other countries" [17]. This population is highly diverse. According to the Migrants Policy Institute, Mexicans were the largest group of US immigrants (23%) in 2022 followed by immigrants from India and China, and other countries, including El Salvador, Vietnam, Cuba, the Dominican Republic, Guatemala, and South Korea

Practitioner Takeaway

- Technical communicators can design digital content and features for websites targeting immigrants and marginalized populations by making them more accessible and inclusive.
 - Considering the functional, sensory, and cognitive affordances in designing and creating navigation tools will help the diverse users of the websites.
 - Technical communicators should design and create website content targeting multicultural users by incorporating assistive technologies that can function in multilingual settings.
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[18]. For the first eight months of 2023, countries from which most refugees came were the Democratic Republic of Congo, Myanmar, and Syria [19]. Between 2012 and 2021, the US granted 280,100 people asylum, with Chinese accounting for the largest group, 23% [19].

In addition to obtaining permanent residency, people need various other types of information regarding their status in the US. Consequently, millions of users will rely on the USCIS website for this information. With this context and grounding, in this article, we present a critical analysis of the USCIS website and argue that the website should incorporate aspects of usability that help and support the diverse needs of immigrants and refugees considering their varied language, literacy, and abilities.

In the following sections, we first review the existing literature on multilingual user interfaces, usability studies, and social justice-centric usability studies as we position our analysis on the interaction of these areas. We then introduce critical interface analysis as our analytical framework and the walk-through approach as our method for conducting the interface analysis of the USCIS site. Our results summarize observations based on functional affordance, sensory affordance, and cognitive affordance, which we will discuss in detail later in this article. Our discussion centers on the takeaways for technical communicators involved in inclusive and socially just interface design for multilingual and diverse users. We conclude by highlighting the limitations of this study and suggesting future work to further advocate for a more socially just UX for linguistically diverse users.

LITERATURE REVIEW

Multilingual Users Interface The US is the only country in the world with more than 40 million foreign-born immigrants, which is one-fifth of the world's immigrants [20]. Most of these immigrants

are in the US legally, and they must navigate the USCIS website [6] at various points in their immigration process.

Mexico, China, India, the Philippines, and El Salvador are the top five countries of birth for these immigrants. In addition, immigrants come from other regions, including Europe, Canada, the Caribbean, other parts of North America, Central America, South America, the Middle East and North Africa, and sub-Saharan Africa [20]. This vast diversity among immigrants is reflected in their cultures, personal preferences, perceptions, and languages. Being born in such a wide array of countries results in differences within this sizable immigrant group in terms of their preferences and perspectives on various matters.

A Pew research project predicts a substantial increase in the number of immigrants in the US, expecting a total rise to 78 million by 2060, accounting for nearly 19% of the country's total population [21]. This increase will lead to greater diversity among the immigrant population. For these diverse immigrants, navigating the USCIS website presents challenges, with language being a crucial factor, alongside cultures, preferences, and perceptions. Immigrants who are not proficient in English or Spanish face particular difficulties, as these are the primary languages used on the website. Despite the availability of multilingual resources in 35 languages, including American Sign Language (ASL) and Braille, the extensive use of mostly English and Spanish creates barriers (see Fig. 1).

Since multilingual users serve as an important group of target users of public service websites, technical and professional communication (TPC) scholars have explored how the interface should be designed to better serve multilingual users although such research is still limited. Almost two-and-a-half decades ago, Sun [22] proposed building a multilingual web design by taking into account the cultural markers of users. She stated:

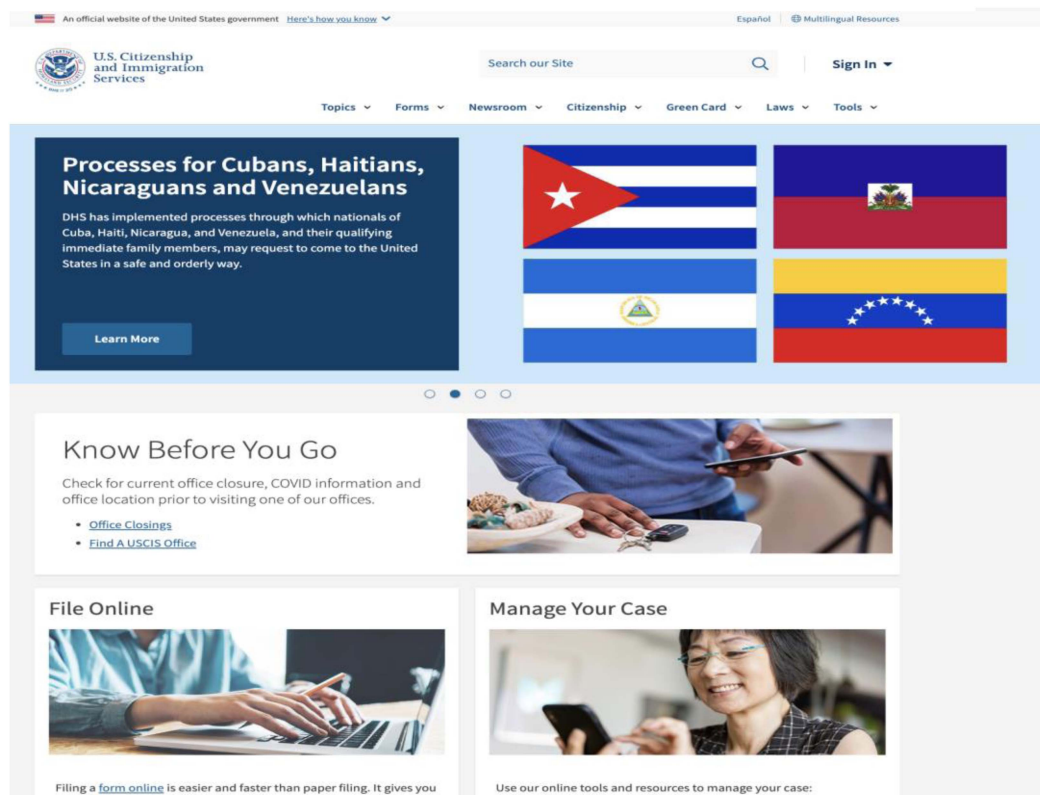


Fig. 1. Screenshot of the USCIS homepage (on 1 December 2023).

Integration of different cultural markers on multilingual Web page design can increase Web usability by making users' navigation more comfortable. [22, p. 101]

Cardinal et al. [23, p. 2] argued that a participatory approach to UX design is needed to engage multilingual audiences as this approach "values the racial, linguistic and cultural diversity of contemporary audiences." They further asserted that when using this inclusive approach, UX designers are required to

incorporate participatory design activities that can help multilingual users to recognize their linguistic and cultural experiences as expertise that can and should guide design decisions. [23, p. 2]

In addition to language, building a multilingual website means accounting for other factors, such as paying attention to image use and visual design, assessing the usability of the designs, and adapting them better to meet multilingual audience needs and expectations [24].

Usability, UX, and Accessibility for Immigrant Users

Usability, as defined by the International

Organization for Standardization [ISO (9241-11: 2018)], refers to

the extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use. [25]

This definition of usability encompasses both the functional requirements and the requirements of users regarding a system, product, or service. Effectiveness and efficiency underscore how the product supports the user's need to reach their goals with accuracy and speed, while satisfaction is entirely determined by the user's perceptions of these qualities [26]. Such requirements and perceptions from users contribute to the next important concept: UX.

The Usability Body of Knowledge defines UX as:

Every aspect of the user's interaction with a product, service, or company makes up the user's perceptions of the whole. User experience design as a discipline is concerned with all the elements that together make up that interface, including layout, visual design, text, brand, sound, and interaction. UX works to coordinate

these elements to allow for the best possible interaction by users. [27]

Our research investigates the holistic interaction experience of immigrant users with the USCIS website interface, with a specific focus on their UX. Although it is impossible to separate usability from UX, we adopt the differentiation proposed by Baxter [28] and a broader process of UX [29]. According to Baxter [28], usability involves more objective and product-based interactions, whereas UX is associated with more subjective and evaluation-based interactions. Nevertheless, we acknowledge that approaches derived from usability studies are viable and effective in understanding UX with an interface because they evaluate how well an interface meets the needs of users, bridging the gap between objective functionality and subjective evaluation.

Scholars in the field of TPC have conducted usability studies focusing on immigrant users, with at least two studies influencing our research. Rose et al. [30] conducted usability studies centered on health insurance information for immigrant patients in a local community health center. Their findings revealed that factors, such as limited English proficiency and health insurance literacy, act as barriers for immigrant patients in enrolling for health insurance [30]. Upon reviewing the methodology of community-based usability studies involving immigrant users, Rose and Racadio [31] argued that a traditional and rigid scenario-driven approach proved ineffective with this user group. Instead, they opted for a more conversational facilitation approach, utilizing cultural and linguistic expertise to tailor usability methods to the specific needs and context of immigrant users [31].

In the TPC field, several scholars, such as Oswal and Hewett [32], Palmeri [33], Salvo [34], Oswal and Melonçon [35], Palmer and Palmer [36], Strantz [37], and Vazquez [38], have stressed the various issues of accessibility. Accessibility, as it relates to disability studies as well as technology, comprises a multidisciplinary scope in the TPC field. This field's current expertise in accessibility stems from several knowledge areas, such as web design, usability, specification design, psychology, and engagement with legal writing [39]. The TPC field, a user-centered applied discipline, advocates for making web content accessible and inclusive to its variety of users, including users who speak languages different from the one in which the content is written.

In her case study on multilingual technical content creation, Gonzales [10] emphasized the significance of treating language access as a crucial aspect of accessibility when producing and disseminating digital research. Our research suggests that apart from the accessibility challenges typically addressed in disability studies, immigrant users of the USCIS website must also contend with unequal language access to effectively comprehend and utilize the website for their benefit.

Social Justice-Centric Usability Studies Our study is driven and informed by social justice-centric usability studies. Jones and Walton stated that social justice research

investigates how communication, broadly defined, can amplify the agency of oppressed people—those who are materially, socially, politically, and/or economically under-resourced. [40, p. 3]

This approach argues for giving agency to the under-resourced and marginalized people who have been oppressed for any reason either materially, socially, culturally, or for any other reasons. The social justice approach is for creating a respectful society as it stands against any form of unequal distribution of power. In this vein, Hass and Eble argue that

social justice approaches are informed by cultural theories and methodologies, but they also explicitly seek to redistribute and reassemble—or otherwise redress—power imbalances that systematically and systemically disenfranchise some stakeholders while privileging others. [41, p. 4]

If the current USCIS website's interface design is guided by certain design principles, it is important to examine whether such design involves power imbalances by favoring certain groups of users more than others.

Considering the increasing emphasis on social justice within the field, scholars in TPC have committed to advancing more equitable, inclusive, accessible, and socially just usability studies. These studies specifically center on marginalized and underrepresented UX. Agboka [42] proposed decolonial approaches as an alternative to modernist approaches to address emerging issues of social justice, culture, and power in post-colonial, unenfranchised/disenfranchised sites. Rose et al. [30], [31] echoed the decolonial approaches and made adaptations in their usability

facilitation approach to address the cultural and educational differences in their immigrant participants and captured a wider range of usability considerations. Working with immigrant and transnational communities, Gonzales argued that

by leveraging technical communication's long-standing emphasis on making information accessible across differences, there is an opportunity for this field to further embrace complexity in working toward justice within global communication practice. [43, p. 8]

Dedicated to advancing linguistic justice in speech technologies, Lawrence asserted that technical communicators should

advocate for a more socially just design of speech devices and ... assess and mitigate potential harms to marginalized communities with varying language backgrounds. [44, p. 1]

Centering on immigrant users as a marginalized and underrepresented group, our research aligns with the effort in social justice-centric usability studies and advocates for language access and inclusive usability studies when designing interfaces for and with multilingual communities.

Looking back to our study context, the USCIS website is navigated by the 46.6 million immigrants who account for 14.6% of the US population as of January 2022 [45] at some point during their legal status in the US. It is important to explore whether the current design of the USCIS website excludes certain users from this large group of diverse immigrants. A social justice approach helps bring a perspective that amplifies agency for those who may be excluded due to linguistic, cultural, political, and other reasons. The interface design of the USCIS website is not just like any other website; more important, we believe that the underlying design logic, which involves power and its redistribution to all users, is of utmost importance. Therefore, we believe that studying the interface design of the USCIS website is important to uncover pertinent issues.

RESEARCH QUESTIONS

We ask and answer the following questions in this article.

1. How does the USCIS website's content cater to diverse immigrant populations in terms of usability, specifically considering Web Content Accessibility Guidelines (WCAG) for accessible websites?

2. From a UX perspective, are the USCIS website navigation tools obtrusive in presenting information? Are there issues of power and privilege through the inclusion/exclusion of certain voices [9]?
3. What ideological and cultural assumptions does its interface design impart to diverse website users through its tools, content organization logic, and visual style?

RESEARCH METHOD

To understand how the USCIS website impacts users' ability to access information and interact with the interface, we conducted a critical interface analysis. This method "layers theory, critique, and reflection" [4, p. 391]. The USCIS website is a web-based interface where one can find information on immigration, citizenship, and related topics. However, first-time users might feel overwhelmed by the variety of information, topics, and numerous buttons on the website, even with a virtual assistant (VA) called Emma.

In our analysis, we considered ourselves both users and researchers of this website, drawing from our own experiences as immigrants in the US. This dual perspective allowed us to incorporate our multilingual and multicultural backgrounds into the analysis. Because our study is based on our personal experiences and not on human-subjects research, it does not fall under that category and did not require Institutional Review Board (IRB) approval. Below, we describe the two analytical methods we employed.

Critical Interface Analysis We adopt the critical interface analysis of Sano-Franchini who argued that

writing and design are epistemic, and that human beings make knowledge and meaning from interpretations—whether conscious or subconscious—of signs, including alphabetic textual and visual design. [4, p. 391]

This way of analysis helped us analyze the usability features of the USCIS website. For a multilingual and multicultural user, knowledge-making abilities vary based on literacy and cultural backgrounds that can impact user decisions and processing information. Thus, the major principles of usability—learnability, memorability, error recovery, utility, and time—guided our deeper examination of the USCIS website.

Stanfill has stated that “Web interfaces similarly structure actions by making some things more possible than others” [46, p. 1060]. The prominence of certain elements in web interfaces indicates their importance and power. That prominence comes with productive tactics. For example, if a user must follow five additional steps to find a specific item compared to another item that is accessible in one step, it indicates the web designer’s unequal prioritization of these items. In addition, there is always an imbalance between the configuration of human activities and the screen-based and device-driven affordances [47]. Hence, in this article, we use critical interface analysis to understand whether users successfully achieve their goals when searching for information on the USCIS website.

Foucault and Gordon’s [48] concept of power as productive, regulatory, and normalizing explains how power incites, encourages, and produces certain practices. Power, at times, represses certain actions and promotes others. Stanfill further contended that

examining what is possible on sites—features, categories of use foreground, and how technological features make certain uses easier or harder—illuminates the norms of use. [46, p. 1061]

The structuring of elements in specific ways reinforces behaviors that align with normativity [49].

When web designers utilize certain interface affordances and develop certain items, features, or categories, they encourage certain behaviors. These behaviors are guided by “cultural common sense about what users do (and should do)” [46, p. 1061]. However, emphasizing one cultural perspective may overlook other practices because not all site visitors navigate the interface the same way. Thus, it is important to consider these diverse practices. The productive power of available affordances and interface should be guided by the target users and the website’s purpose.

Walkthrough Method Specifically, we used the walkthrough method as a step-by-step approach to guide us through the data documentation, note-taking, and analysis process [50]. Light, Burgess, and Duguay proposed the walkthrough method as a way of

embedded cultural references to understand how it guides users and shapes their experiences. [49, p. 882]

This method has been widely used to analyze interfaces on popular apps, such as TikTok, Douyin, and Momo [51], [52], [53]. We adopted this method as it allowed us to conduct a nuanced usability analysis of the website, focusing on various aspects, such as culture, documentation, representation, and more.

Despite its typical use for analyzing apps, we considered this method appropriate for us to adopt and adapt for analyzing the USCIS’s website interface for the following reasons. First, it involves researchers’ participation, observation, interaction, documentation, and interpretation of an app’s screens, features, and flows of activity to understand how it guides users, allowing us to conduct usability analysis. As an observational and interpretive method, it aligns with our purpose to position ourselves as users to critically analyze various features and accessibility of the interface of the USCIS website.

Second, its main distinction from other usability testing and interface analysis methods lies in its interpretive power underpinned by theoretical frameworks in science and technology studies and cultural studies [50, p. 882]. As argued by Light, Burgess, and Duguay, the walkthrough method is not to test whether

users respond to an interface in the ways its designers intended but rather to illuminate the material traces of those intentions, and thereby to critically examine the workings of an app as a sociotechnical artifact. [50, p. 886]

This objective resonates well with our research goals as we are looking for not only the technological mechanisms of the interface of the USCIS website but also how such interface artifacts constitute social reality and power relations [54].

Third, the emphasis on analyzing “everyday use” in the walkthrough method offers us the data-collection approach that requires us as researchers to engage closely with the interface. Using a step-by-step walkthrough technique, we progressed through the requirements, screens, features, and activities of the interface to understand and interpret

engaging directly with an app’s interface to examine its technological mechanisms and

its embedded sociocultural representations as much as its technological features or data

TABLE I
THREE CATEGORIES OF AFFORDANCES

Affordance Type	Description	Example
Functional Affordance	Includes the design features that help users accomplish certain work [55]	The USCIS website allows users to download forms and file their cases online.
Sensory Affordance	Relates to visibility, legibility, and/or audibility of the site, or the aesthetics consideration of the site [46]	The USCIS website has a flash banner toward the top of the landing page, which informs site users of the latest happenings at the USCIS.
Cognitive (Including Language) Affordance	Helps, aids, supports, facilitates, or enables thinking and knowing about something.	The USCIS website has multilingual resources that offer content in multiple languages to aid non-English speaking users in learning about information.

outputs, which also have social and cultural influences. [50, p. 885].

Based on the critical interface analysis and the walkthrough method, we developed three categories that we explain in detail in our results and discussion section.

RESULTS AND DISCUSSION

We followed the walkthrough method [50] in combination with the critical interface analysis [4]. By assuming the role of users with specific immigration objectives, we thoroughly examined the steps and content related to green card applications on the USCIS website. We chose to focus on green card applications as this is a high-priority need for most immigrant users of the USCIS website. Our research findings and reflections have been meticulously recorded to address the research questions.

We organized our findings based on our observations concerning three categories of affordances (see Table I) when interacting and engaging with the website to achieve our objectives. These categories had their roots in Harston's [55] and Stanfill's [46] studies, with our addition of language to the category of cognitive affordance to reflect language as an important website interface feature. When presenting our findings on a particular affordance category, we prioritized that category while acknowledging the interrelated nature of all affordance categories as they are intertwined and integrated into the design [55].

Functional Affordance Following Hartson's [55] definition, we define functional affordance as the website functionalities that allow specific user groups to achieve goal-oriented actions. For example, with electronic forms available on USCIS, users who need to apply online can download appropriate forms and complete those tasks. These forms represent a functional affordance of the website's online application.

To perform the analysis of USCIS's functional affordance, we turn to the hierarchical and decomposable nature and the relational inherence of functional affordance as argued by Savoli [56]. Functional affordance is hierarchical and decomposable in that "the affordances can be categorized as being in higher or lower levels in terms of their means-ends sequences" [56, p. 26]. For example, a functional affordance at its highest level on the landing page of USCIS is "File Online." This functional affordance can be decomposed into at least three sublevels, including filing a form online, creating an account, and watching how-to videos.

Functional affordance is also relational as the outcome of an affordance is dependent on both users and the functional and design features of an interface. For example, the landing page of USCIS has an interactive VA "Ask Emma" who can help answer site users' questions. However, based on the description of "Ask Emma," Emma cannot respond to user's concerns precisely when users ask in limited terms such as "renewal," "visa," or "appointments." Instead, Emma asks users to use complete sentences saying, "Please write complete statements" and offers a lengthy example of how to

ask in complete sentences in English or Spanish. Given the varied levels of English and Spanish language competencies of users, this means that only users who can write complete English or Spanish sentences might find interacting with Emma useful. Thus, a user's proficiency in English or Spanish significantly influences their experience and the outcome of their interaction with Emma [6], [20].

To improve this limitation, USCIS website designers should clarify instructions for chatting with Emma so that users with limited English or Spanish proficiency are not deterred from using the assistant. Instead of "Please write complete statements," the instructions should be more like "Try to make your questions short and specific."

By analyzing the hierarchical structure of the landing page of USCIS, we identify the following three higher level functional affordances and their subfunctions (see Table I), which also construct norms. The landing page affords the most prominent function of acquiring and accessing information. Information acquisition can be achieved by browsing the webpage and locating the information, using the "search bar" at the top of the webpage, asking "Emma," a chatbot floating in the right corner at the top of the webpage; searching in "tools and resources" toward the bottom-right of the webpage, or following the social media at the bottom of the webpage.

The landing page affords the second function of "filing online," which is nestled in the middle of the page. Since this function represents multiple complex functions, it can be decomposed into three subfunctions to complete each task. It is composed of at least three basic subfunctions "how-to video," "create an online account," and "fill out the forms."

The final prominent function the landing page affords is "manage your case," which allows users who have submitted their application to follow up on the case and trace the process. This function can be decomposed into three subfunctions "checking case status," "checking process time," and "updating information."

The landing page also affords interactivity between the users and the content. Such interactivity is mainly provided by the chatbot, Emma, a random invitation for site use evaluation, and social media platforms, such as Facebook, Twitter, YouTube, Instagram, LinkedIn, and Email. Notably, there can

be a legitimate reason for limiting such interaction to individual levels like the user's right to privacy.

However, it is important to consider why communal action might sometimes be preferable to individual action in the context of immigration services. Although official information is crucial, a moderated community forum could help clarify common misunderstandings and share tips for navigating processes. In addition, communal discussions could help USCIS identify recurring concerns or confusing aspects of their processes, thus informing improvements to their services and communications.

Although maintaining the integrity and security of immigration processes is critical, incorporating some level of moderated community interaction could enhance the UX, improve information accessibility, and potentially reduce the load on official support channels. Any community features would need to be carefully designed to protect privacy, prevent the spread of misinformation, and clearly distinguish between official information and community-contributed content. By striking this balance, USCIS could leverage the benefits of communal action while upholding its standards as a government agency.

After identifying the possible and visible functional affordances, we now highlight the norms these functional affordances have structured. This website normalizes English or Spanish proficiency as the prerequisite to using the functions provided. We come to this finding for two reasons. First, Emma suggests lengthy examples of how to ask questions in complete English or Spanish sentences. Hence, users are expected to have a good command of English or Spanish proficiency to acquire information on the website. However, Emma's instructions incorrectly imply that more language proficiency is required than is the case. This fact might deter users from trying to communicate with Emma. The instructions for chatting with Emma should therefore be revised to explain Emma's abilities more precisely.

In addition, the resources for multilingual users to access information are very inconspicuous because users must know English or Spanish because all the landing pages or tools are written in these two languages (see Fig. 1). A website in only two languages favors users who can easily use English or Spanish rather than providing multilingual tools for translation and help. Because most immigrants

are from Mexico, China, India, the Philippines, and El Salvador, it is surprising that the site is not also fully available in simplified Chinese, Hindi, and Filipino. In addition, if USCIS is to avoid impediments to Arabic and African immigrants, the site should also be fully available in Arabic, Swahili, French, Portuguese, Hausa, and Yoruba.

This website normalizes that users have adequate knowledge to use it for the first time. In other words, for first-time users, functions like site navigation, frequently asked questions, and how-to videos are not immediately available and visible. Palmer and Palmer, for instance, recommend four main principles of WCAG to enhance the accessibility concern on any website which include that “the information needs to be (a) perceivable, (b) operable, (c) understandable, and (d) robust” [36, p. 402]. As a user, it is important whether the USCIS web information is at the level of its diverse users who come from varied linguistic, cultural, and sociopolitical backgrounds. The USCIS website should be able to treat its diverse users based on their needs rather than functioning as a platform serving those who have a high command of English or Spanish and come from certain cultural backgrounds.

Making a dynamic website depends on how accessible its content is to its users. To make any webpage accessible, several factors play significant roles. For instance, if any website’s users are diverse in terms of language, culture, and shared knowledge, web designers should pay attention to those elements to accommodate diverse groups of users. A vast number of immigrants mean a vast number of differences and consequently a higher possibility of encountering problems accessing any web content easily. In addition, the various forms of the disabled population mean that web accessibility will inevitably be an additional stumbling block.

Although accommodating all sorts of users with all types of needs can pose a challenge, designing a webpage should at least avoid a monomodal approach. The website should guarantee to accommodate its users with more than one ability. Palmer and Palmer mentioned that

these accommodations can take many forms, but what they all have in common is that they enable access to information in ways that do not depend on a single sense or ability. [36, p. 400]

Even though the USCIS site is available in Spanish and English, with additional multilingual resources

available in 35 languages, including ASL and Braille, it still does not adequately serve the heterogeneous immigrant users of this website. Its current design requires users to know either English or Spanish because all the navigation tools rely on these languages to access the multilingual resources. In this way, instead of treating the entire immigrant population as members of one or two groups, the USCIS webpage designers should consider various factors, such as differently able immigrant users, and users of multiple languages, cultures, and shared knowledge.

Sensory Affordance Hartson defined sensory affordance as a “design feature that helps, aids, supports, facilitates, or enables the user in sensing (e.g., seeing, hearing, feeling) something” [55, p. 322]. In essence, sensory affordance encompasses “design features or devices associated with visual, auditory, haptic, tactile, or other sensations” [55, p. 302]. As one of the important dimensions of affordance, sensory affordance plays a crucial role in enhancing emotional participation and reducing the psychological distance between users and artificial intelligence (AI)-assisted services [57]. Scholars in the field of virtual and interactive agents argued that sensory affordances help “reinforce other perceived affordances” [58, p. 88].

During our analysis of the USCIS interface, we sought to identify elements or tools that represent sensory affordance. We examined multimodal elements, such as interactive virtual tools like Emma, videos, and images that can immediately engage users’ senses and influence their perceptions and attitudes. Despite the effort of USCIS to leverage Latina identity as a strategic design choice for integrating the VA Emma into its website [20], our findings revealed that USCIS’s website is less helpful and useful to immigrants who prefer interactive engagement with the interface using languages other than English and Spanish. This conclusion was based on the following observations.

First, Emma, as the most prominent and immediate interactive tool, has significant imitations as it can interact only in English and Spanish and requires users to input appropriate terminologies to access information. Consequently, individuals who are not proficient in these two languages may feel excluded from utilizing Emma’s services. The Pew Research Center identifies countries, such as Mexico, China, and India, as top birthplaces for immigrants in the US [21]. Given the diverse US immigrant population with varying

linguistic backgrounds and English proficiency, Emma's restriction to English and Spanish creates a psychological barrier for users whose primary languages are neither English nor Spanish.

Although English has been used in schools in countries like India and China, the users in these countries have their languages, and psychologically, English still creates barriers for these users. The initial interaction with Emma might exclude non-English or Spanish speakers, as they do not perceive Emma as an effective bridge between their language and USCIS services. Furthermore, this feeling of distance between Emma and non-English and Spanish speaker users is supported by researchers' argument that "the Emma interface makes normative claims about citizenship and inclusion to 'hail' Latinx users as ideal citizens" [59, p. 909].

Second, videos, as a popular multimedia form for users to learn about new objects and processes, are limited and underdeveloped on the USCIS website. As mentioned earlier, there are two spots on the landing page of the website to access videos: one is embedded under the column "File Online," and the other is the YouTube icon at the bottom-left of the website. However, their visibility remains limited and underdeveloped. Consequently, first-time users may struggle to identify and utilize this communication channel effectively. Our investigation into the green card application process led us to search for specific instructional videos on the USCIS website, employing both the site's search bar and the official USCIS YouTube channel. Regrettably, the results yielded no videos instructing users about the essential aspects of applying for a green card. The majority of available "how to" videos produced by USCIS cater to naturalization, citizenship, online form filling, and E-verify, leaving aspiring green card applicants with a scarcity of video resources to guide them through the entire process.

Users at different stages of their green card application may request varied information types and channels to guide them through the process. For users who are in the early stage of the green card application and need a dynamic and multimodal type of information to quickly understand the process, videos might be one of the effective communication channels. This inadequacy in video resources regarding green card applications highlights a crucial area for improvement in enhancing UX and accessibility for USCIS website visitors.



Green Card Eligibility Categories

Find the eligibility category that fits your immigration situation, research how to apply, and learn whether your family members can also apply with you.

Fig. 2. Image about green card eligibility categories (see <https://www.uscis.gov/green-card>).

Third, the USCIS website's utilization of images as visual aids and content occasionally presents discrepancies with the accompanying text, leading to confusion and exclusion among users. Li et al. [60] have demonstrated the significant impact that images can have on consumers' sentiment and perception of the usefulness and enjoyment of online reviews. Likewise, images that convey rich and vivid information can enhance users' comprehension of textual content. Conversely, when images are incongruent with the accompanying text, they may create a sense of detachment among users.

In our analysis, we observed certain images in the "Topics" section under "Green Card" that exhibited dissonance either with the text or its intended implications. For instance, Fig. 2 displays an image alongside the description for the topic of "Green Card Eligibility Categories." As individuals with Asian cultural and birth origin backgrounds, we found it challenging to identify with any implied ethnicities conveyed by the image. Such images that only partially feature the ethnicities of the immigrant population may cause those who are not represented in the visual to feel underrepresented and excluded.

Similarly, Fig. 3 portrays an image related to the topic "Replace Your Green Card." However, the



Fig. 3. Image about “Replace Your Green Card” (see <https://www.uscis.gov/green-card>).

image, depicting a person smiling while holding an American national flag in each hand, lacks a direct association with the topic of replacing one’s green card. These disparities between text and images on the USCIS website underscore the need for greater attention to visual content coherence and cultural inclusivity. Addressing these concerns can foster a more inclusive and user-friendly experience for all visitors, regardless of their cultural backgrounds or ethnicities.

Cognitive Affordance According to Hartson, “a cognitive affordance is a design feature that helps, aids, supports, facilitates, or enables thinking and/or knowing about something” [55, p. 319]. When applying the lens of affordances to conducting interface analysis, Stanfill expanded this definition and argued that “cognitive affordances are perhaps the most expensive, encompassing linguistic and nominative features of sites” [46, p. 1068]. In our analysis of the USCIS’s website interface regarding cognitive affordances, we primarily concentrated on the textual content and tone that are embedded in the interface.

Studies have identified language as a significant concern for multilingual individuals when accessing web content. A study by Vazquez [38] highlighted three major difficulties encountered by multilingual users: language selection, untranslated content, and user-side technology issues. In his study, Obsborn [61] showed a case of how linguistic dimension functions as an important factor in the digital divide in Africa. Like other

technical aspects of the divide, such as connectivity and access, choice of language (orthographic representation of African languages in information communication and technology) and literacy issues play equal roles in multilingual contexts across Africa. Villa-Nicholas and Sweeney analyzed the interface of the USCIS website, especially the role of Emma, a VA technology in terms of providing cultural affordances to Latinx users, and concluded its positive impact as it provided “an assimilated citizen-consumer” to its users [59, p. 909]. Although designing a website to address this enormously vast immigrant population in the US context can be extremely daunting, paying attention to the various factors that scholars [22], [38], [61], [59] have suggested can be a useful strategy to make the USCIS website more accessible and user-friendly.

Our first observation was concerned with the use of the disturbing term “alien” to refer to immigrant users. Although the US recognizes itself as “a nation of welcome,” the use of the term “alien” to officially describe individuals and corporations from foreign lands has long been a political strategy (see Fig. 4). This word has its Latin origin in “alienus,” meaning a stranger, a foreigner, or someone not related by blood. In the modern context, regarding pop culture, the word “alien” is often used to describe imaginative creatures from other worlds. In contemporary times, using “alien” as an official term to describe immigrants in a country with a historical foundation of immigrants sounds discriminatory, even though it might simply mean someone belonging to a foreign country. According to the *Merriam-Webster Dictionary*, the use of the term “alien” is stronger than foreign in suggesting opposition, repugnance, or irreconcilability” [62].

Another literal meaning of “alien” is extraterrestrial. This harsh terminological usage not only creates confusion but also exemplifies how immigrants are treated legally and as a political strategy. Although other immigrant-welcoming countries like Canada use the term “foreign national” instead of “alien,” Australia used to refer to immigrants as “alien” until 1983; they have since changed to “non-citizen.” The use of “alien” is not immigrant-friendly; instead, it reflects a deeply rooted attitude toward this minority group. If the US defines itself as “a nation of welcome and possibility of fairness, integrity, and respect for all,” labeling any minority group as “alien” is ambivalent [6]. Michael Lechuga, Assistant Professor at the University of New Mexico, mentions that



The screenshot shows the top portion of USCIS Form I-140. At the top left is the Department of Homeland Security seal. The title 'Immigrant Petition for Alien Workers' is centered. Below it, 'Department of Homeland Security' and 'U.S. Citizenship and Immigration Services' are listed. On the right, 'USCIS Form I-140' is displayed, along with 'OMB No. 1615-0015' and 'Expires 03/31/2024'. The form is divided into several sections: 'For USCIS Use Only' (a vertical column), 'Fee Stamp', 'Priority Date', 'Consulate', and 'Action Block'. Below these are 'Classification' and 'Certification' sections, each with multiple checkboxes for different categories of workers. A 'Remarks' section is at the bottom.

For USCIS Use Only	Fee Stamp	Priority Date	Consulate	Action Block
	Classification ☐ 203(b)(1)(A) Alien of Extraordinary Ability ☐ 203(b)(1)(B) Outstanding Professor or Researcher ☐ 203(b)(1)(C) Multinational Executive or Manager ☐ 203(b)(2) Member of Professions with Advanced Degree/Exceptional Ability ☐ 203(b)(3)(A)(i) Skilled Worker ☐ 203(b)(3)(A)(ii) Professional ☐ 203(b)(3)(A)(iii) Other Worker		Certification ☐ National Interest Waiver (NIW) ☐ Schedule A, Group I ☐ Schedule A, Group II	
Remarks				

Fig. 4. Screenshot of Form I-140 titled “Immigrant Petition for Alien Workers.”

The words we choose to use for one another have an impact on the way people are perceived and thus treated in our society. Dehumanizing words have no place in public discourse. [63]

This kind of word choice has significant implications for how the country seeks to segregate the minority population. It involves leveraging immigrants’ extraordinary talents while rejecting their identities.

Furthermore, the use of terms like “alien” and “illegal” is the result of a perpetual colonial mindset. It further debunks the idea that

in many ways, relying on a dehumanizing term like “alien” allows political figures to erase the experiences of migrants who are fleeing nationalist violence, climate catastrophe and economic despair. [63]

The “illegal alien” mostly refers to “Latinx/Mestizx and Asian communities in the country who have entered the US without authorization or who have overstayed their visa” [63]. However, the use of such terms is aggressive to a specific minority community and sounds biased. For example, one news article on the USCIS website in 2014 is entitled “3 illegal aliens charged with immigration violations” [64], and another in 2017 is entitled “Illegal alien indicted for hampering with ankle monitor” [65].

In response to widespread criticism, President Biden issued a memo to the nation’s two major immigration enforcement agencies, Customs and Border Protection and Immigration and Customs Enforcement, instructing them to use the terms, “non-citizen” and “migrant” instead of “alien” on April 19, 2021 [66]. The leader of his opposing

political party immediately criticized President Biden for his move. This also indicates that using these terms remains a political strategy in the nation. However, immediately after Donald Trump returned to office for a second term as President of the United States, he signed several executive orders on his first day in office. Unlike his predecessor, President Biden, Trump promptly reinstated the term “alien” to refer to “immigrants” [69]. Following this directive, the acting secretary of Homeland Security issued an internal memo instructing the use of the term “alien” instead of “immigrants” [70], [71].

From a social justice perspective, if we want action that leads to inclusion, we must recognize that using such terms is not fair to those from minority groups [9]. The use of such terms has indeed been a political strategy in the US, and at times reflects White supremacy, by distancing and labeling immigrant populations as alien. National Public Radio (NPR) reports that

today, “alien” is frequently used by conservative politicians and pundits. The term “alien” and especially “illegal alien” have become so politically charged. [67]

NPR further describes that although this term has been used historically since the 1970s, the use of the term, “alien” has ceased to be a neutral term. Instead, it has become a coded form of bigotry, enabling a certain group of people to be targeted. This also entails a message that “immigrants are not us” and “immigrants are different” despite their lawful presence in this country. We consider that using such legal terms is also a misuse of power, positionality, and privilege.

Being a leading country in science, innovation, advancement, and opportunities for employment, the US is an attractive destination country for millions of immigrants from across the world. Lawmakers and political leaders in the US have the authority to amend legal documents according to their preferences, demonstrating their power and privilege. However, insensitive terms like “alien” to refer to immigrants raise concerns about how they exercise this authority. It can be difficult for the USCIS website designers to change the word “alien” without creating confusion if USCIS and other US government agencies are still using this legal term. The website designers could, however, at least note that “alien” is a historical legal term, acknowledge that the word has acquired negative associations, report efforts to change it, and regret using it.

Furthermore, during our examination of the flash banner section, we encountered a banner titled “July 19: Don’t be a Victim of an Immigration Scam.” Although the message’s intention is undoubtedly genuine, it inadvertently reinforces the perception that immigrants are commonly targeted by scams, which may inadvertently perpetuate negative stereotypes. The framing of this banner title also entails that this population is precariously vulnerable. We believe that the framing of such banner titles can be carefully projected in important spaces like the USCIS website.

Our detailed check on the Chinese translation of the Green Card information also revealed an inaccurate match between Chinese and English. For example, the translation for the item “How Do I Become a Permanent Resident? (Chinese)” is “How do I become a permanent resident as a refugee or political asylee?” (see Fig. 5). That translation is inaccurate and misleading to users who want to find the possible routes to becoming a permanent resident or the categories of eligibility for a green card. In addition, that translation was dated August 2008 and might contain out-of-date information.

Despite USCIS’s efforts to provide multilingual resources to aid diversified immigrants in navigating their webpages, its multilingual resources are scarce, inaccurate, and underdeveloped. For example, since two of us are Nepali, we researched whether there is Green Card information in Nepali under Multilingual Resources. Unfortunately, no information on Green Cards was found in Nepali. Since one of us is from mainland China, we found the Green Card



Fig. 5. Chinese translation of “How Do I Become a Permanent Resident?” (see https://www.uscis.gov/sites/default/files/document/guides/D3_v.1_TChinese.pdf).

information in Chinese. We were able to see 20 items in varied versions of Chinese (see Fig. 6). Most of the items were in traditional Chinese, which is mainly used by Taiwanese, while items regarding EB-5 Immigrant Investor were in simplified Chinese. Although we are unsure about the reasons why the USCIS translated different types of Green Card information into different versions of Chinese, one assumption might be that most immigrant investors might come from mainland China who need to read simplified instead of traditional Chinese. However, given the much greater number of immigrants from mainland China than from Taiwan, translating the Green Card information into traditional Chinese risks alienating immigrants from mainland China. We are not sure how such translation decisions are made, but they may reflect the ideology and stance of the US on closer ties with Taiwan.

The Multilingual Resource Center faces a shortage of documents centering on its diverse users. For example, we were surprised to discover that there is only one document entitled “Information Guide for Prospective Asylum Applicants” and another press release on “DHS Publishes Fair and Humane Public Charge Rule” in the Nepali language.

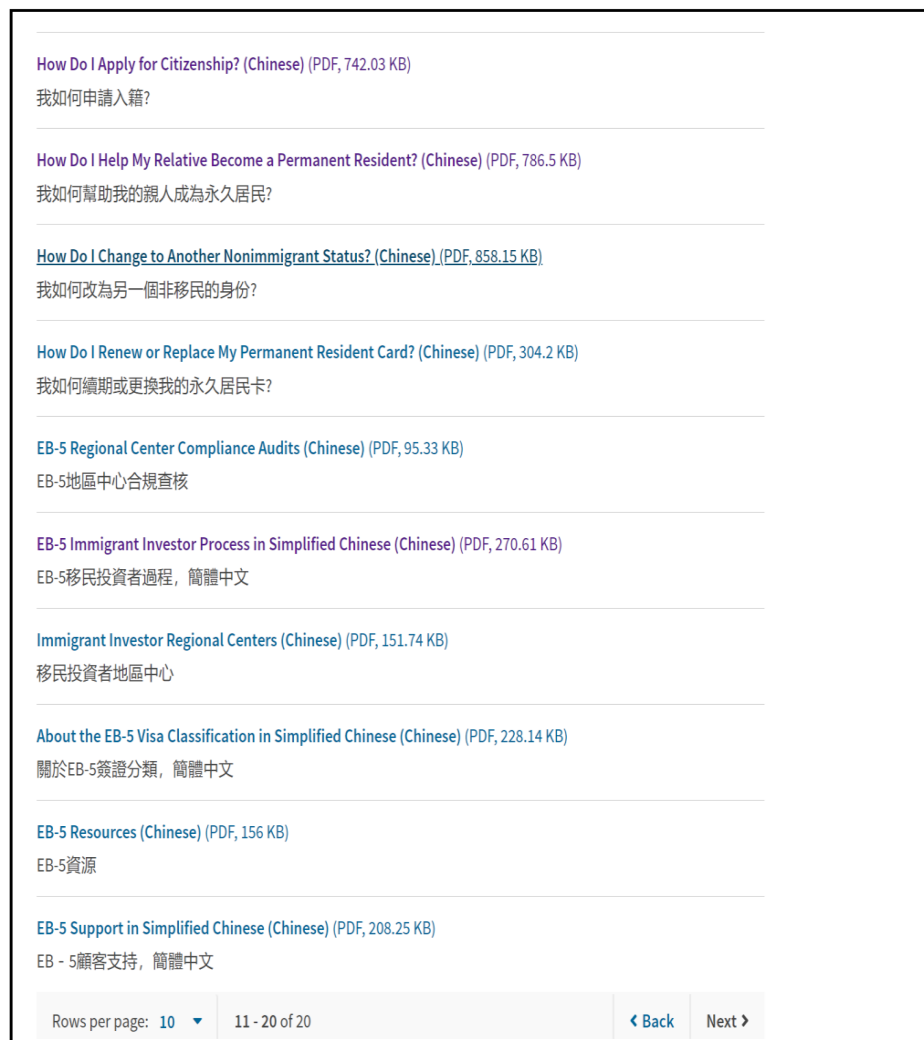


Fig. 6. Green card information in Chinese (see [https://www.uscis.gov/tools/multilingual-resource-center?sel_lang=17502&topic_id\[0\]=33605&items_per_page=10&page=1](https://www.uscis.gov/tools/multilingual-resource-center?sel_lang=17502&topic_id[0]=33605&items_per_page=10&page=1)).

Besides asylum-seeking, other popular pathways for the Nepali-speaking population in the US include the Electronic Diversity Visa, employment-based green card, Bhutanese refugees, and green card through family ties. Unfortunately, the diverse Nepali-speaking population does not find this resource helpful due to the lack of supportive documents. A significant number of Nepali-speaking Bhutanese individuals have been migrating to the US since 2008. As of 2017, more than 800,000 Nepali-speaking Bhutanese have migrated to the US and now reside in 42 states [68]. In addition, two of the authors of this study, who are Nepali-speaking, did not find any of these Nepali-translated documents useful during our green card processes as we adopted employment-based pathways.

Our analysis of the cognitive affordances on the USCIS's website interface underscores the

importance of thoughtful and inclusive design choices. The very limited number of languages incorporated in the multilingual resource section, such as Arabic, traditional Chinese, Russian, Dari, and so forth, may not correctly represent the immigrant population that uses the USCIS website each year.

Offering information in multiple languages is undoubtedly time- and cost-intensive, but it is beneficial for accessibility and inclusivity. For example, although the US provides information in 20 languages for visa applications at ais.usvisa-info.com, visitors are often redirected to English-only pages. This fact creates barriers, even for processes like tourist visas, where the US has a vested interest in being a welcoming nation.

US Consulate pages sometimes offer non-English versions of their information, but they often default

to colonial languages, such as French and Portuguese, rather than the local languages of African countries (e.g., ci.usembassy.gov or ao.usembassy.gov). This lack of linguistic diversity highlights the need for greater investment in the translation and integration of visa information across all US websites that serve as contact points for immigrants.

The use of respectful terminology and accurate translation can significantly impact immigrants' experiences while navigating the website. Addressing these concerns can contribute to fostering a more welcoming and accommodating environment for immigrants seeking information and services from the USCIS. Unless the USCIS changes its web design features, it will continue to function as one of the strong factors of the digital divide. Like other technologies, such as computers and the internet [61], the current mode of USCIS web design can function as an important factor in the uneven distribution of the digital divide.

CONCLUSION

This study provides some insights into how the content and design of the USCIS website can be more usable and inclusive to its heterogeneous immigrant populations. In this study, we recommended several strategies to make the navigation tools of the website more obvious. Instead of being guided by any ideology or making its content a political strategy, the USCIS should pay attention to functional, sensory, and cognitive affordances to make its tools, content organization logic, and visual styles more suitable, useful, and helpful to its diverse website users. Although the users of this website are multilingual, representing various cultural groups, the designers of the USCIS website cannot make insensitive choices of any languages and designs. Instead, the USCIS should attend to its diverse users and their preferences to navigate the site. Although some of the multilingual resources and tools are limited in number and to specific audiences, USCIS should be able to embrace the recent emergence of AI on the site. For instance, the USCIS can embed AI-assisted translation tools in Emma to facilitate multilingual communication.

Limitations Although the recommendations provided in this article can help USCIS make its web content and design more inclusive and inspired by the social justice approach, it is also important to acknowledge that this study has several limitations.

First, instead of recruiting any particular number of research participants and expanding the breadth of the study, we believed that it is equally important to focus on the depth of the study by triangulating our multilingual experiences as researchers with our data-analysis process. However, it can always be helpful to expand the breadth of the study by including a certain number of research participants and incorporating their perspectives into the study. As we continued delving into this project, our research team expanded the data-analysis process utilizing previous studies with our multilingual website navigation experiences of the USCIS website. In the future, we encourage technical communicators to expand this study by including diverse sets of research participants and include their perspectives to triangulate with the other data sources.

Furthermore, although all of us were engaged in a series of dialogues and the collaborative nature of the data-analysis process by embedding our findings with each of our lived experiences, drawing conclusions by solely relying on one single project's findings has its limitations. Thus, in future projects, it would be helpful to follow a similar study by recruiting research participants that reflect the current ratio of immigrant populations who encounter the USCIS website. Our team is pursuing a follow-up project that focuses on multifaceted aspects of the USCIS website's interface design.

Suggestions for Future Research As researchers continue expanding the findings and recommendations of this project, it would have been useful to conduct multiple usability testing sessions for multilingual immigrant users. Conducting such testing would not only help incorporate the voices and lived experiences of those users but also strengthen and enhance the impact of the study to bring some real changes in the interface design of the USCIS website. Overall, the recommendations shared in this study are just food for thought to reconsider the social justice approach, adopt an inclusive approach in designing the USCIS website, and serve larger heterogeneous groups of immigrant populations.

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Although we copy-edited this article during the week when President Trump returned to the White House, it was originally written during the Biden administration in 2023 and provides an analysis of the website during that period.

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